

Last time

- Contraction maps
- Linear stability analysis

We derived the evolution of infinitesimal perturbations to the state:

$$\frac{du(t)}{dt} = \nabla V(\varphi^t(x)) u(t)$$

$$\text{Flow : } \frac{d\varphi^t(x)}{dt} = V(\varphi^t(x))$$

$$\varphi^t(x) \in \mathbb{R}^n.$$

$u(t)$: is an infinitesimal perturbation to $\varphi^t(x)$.
 $\in \mathbb{R}^n$

$\{\varphi^t(x)\}_{t \in \mathbb{R}^+}$: orbit of x .

Specialize to fixed points:

$$\varphi^t(x^*) = x^* \quad \forall t$$

x^* is called a fixed point of flow φ^t .

$$\begin{aligned} \frac{du(t)}{dt} &= \nabla V(\varphi^t(x^*)) u(t) \\ &= \nabla V(x^*) u(t) \end{aligned}$$

Solution

$$u(t) = e^{t \nabla V(x^*)} u(0)$$

$A \equiv \nabla V(x^*)$ Jacobian matrix of $V: \mathbb{R}^n \rightarrow \mathbb{R}^n$ at fixed point x^* .

$$u(t) = e^{tA} u(0)$$

if $\text{Re}(\lambda) < 0$ for all eigs λ of A , then,

$$\|u(t)\| \leq C e^{-\alpha t} \|u(0)\|$$

for some $\alpha > 0$, $C > 0$.

if at least one eigenvalue ^{of A} has Re part > 0 , then, for "almost all" $u(0)$
 $\|u(t)\| \rightarrow \infty$ as $t \rightarrow \infty$.

Why?
$$\begin{aligned} u(t) &= e^{tA} u(0) \\ &= P^{-1} e^{t\Lambda} P u(0) \end{aligned}$$

($\because P^{-1} \Lambda P = A$ is the eigendecomposition)

A fixed point is a hyperbolic fixed point when $\text{Re}(\lambda) \neq 0$ for any eigenvalue λ of A .

Aside on maps

$$x_{t+1} = F(x_t) \quad \text{map}$$

$$u_{t+1} = \nabla F(x^*) u_t \quad \text{linearization}$$

$$u_t = (\nabla F(x^*))^t u_0$$

$\text{Re}(\lambda) > 1$ even for one eigenvalue λ of $\nabla F(x^*)$. $\rightarrow$ almost all solns u_t blow up

e.g. if u_0 is an eigenvector
 $\nabla F(x^*) u_0 = \lambda u_0$ for

$$\lambda < 1. \quad u_t = \lambda^t u_0$$

Ex: Power iteration from linear algebra.

Why study evolution of infinitesimal perturbation?

The behavior of the nonlinear system is described "locally" by its linearization

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Asymptotic stability: We say that a fixed point x^* ($\varphi^t(x^*) = x^*$) is asymptotically stable if there exists a neighborhood U of x^* such that $\varphi^t(x) \rightarrow x^*$ as $t \rightarrow \infty$ for any $x \in U$.



Linear stability: $\operatorname{Re}(\lambda) < 0$ for any eigenvalue λ of $A \equiv \nabla V(x^*)$.

Theorem

Linear stability implies asymptotic stability

Assumptions on V : $V \in C^1$
and $\exists x^*$ s.t. $V(x^*) = 0$.
 $\downarrow$
then exists

and $A \equiv \nabla V(x^*)$ has all stable eigenvalues i.e. $\operatorname{Re}(\lambda) < 0$ for all eigenvalues λ of A .

Proof

Given: $\frac{d\phi^t(x)}{dt} = V(\phi^t(x))$

(Consider x^* of ϕ^t)

That is $\phi^t(x^*) = x^* \Rightarrow V(x^*) = 0$.

Let $y(t) = \phi^t(x) - x^*$

(should be easier to derive evolution of $y(t)$).

$$\frac{dy(t)}{dt} = V(\phi^t(x)) = V(y(t) + x^*)$$

$$= V(x^*) + \nabla V(x^*)y(t) + N(y(t))$$

(Take $N(y(t)) = V(y(t) + x^*) - \nabla V(x^*)y(t)$)

$$\frac{dy(t)}{dt} = \underbrace{\nabla V(x^*)}_A y(t) + N(y(t))$$

if $V \in C^1(\mathbb{R}^n)$, then,

$$N(y(t)) = o(y(t))$$

from Taylor expansion

$$\frac{dy(t)}{dt} = A y(t) + N(y(t))$$

$$y(t) = \underbrace{e^{tA}}_0 y(0) + \underbrace{\int_0^t e^{(t-\tau)A} N(y(\tau)) d\tau}_0$$

(check that this is a soln

$$\frac{dy(t)}{dt} = \boxed{A e^{tA} y(0)} + \boxed{A e^{tA} \int_0^t e^{-\tau A} N(y(\tau)) d\tau}$$

$$+ \cancel{e^{tA}} \cancel{e^{-tA}} N(y(t))$$

$$\frac{dy(t)}{dt} = A y(t) + N(y(t))$$

Recall for any $y(0)$, we have that $\|e^{tA} y(0)\| \leq C e^{-\alpha t} \|y(0)\|$

Putting this in our solution

$$y(t) = e^{tA} y(0) + \int_0^t e^{(t-\tau)A} N(y(\tau)) d\tau$$

$$\|y(t)\| \leq \|e^{tA} y(0)\| + \int_0^t \|e^{(t-\tau)A} N(y(\tau))\| d\tau$$

$$\leq C e^{-\alpha t} \|y(0)\| + C \int_0^t e^{-\alpha(t-\tau)} \|N(y(\tau))\| d\tau$$

For any $K > 0$,

$$\exists \delta > 0 \text{ s.t. whenever } \|y(0)\| < \delta,$$

$$\|N(y(\tau))\| \leq K \|y(\tau)\|$$

(differentiability of V and continuity of $t \rightarrow y(t)$)

$$\rightarrow \text{assumption on } y(0) = \phi^0(x) - x^* = x - x^*$$

$$\|y(t)\| \leq C e^{-\alpha t} \|y(0)\| + CK \int_0^t e^{-\alpha(t-\tau)} \|y(\tau)\| d\tau \rightarrow \textcircled{A}$$

Gronwall's inequality

$$\beta(t) \leq \left[c + \int_0^t r(\tau) \beta(\tau) d\tau \right]$$

then,

$$\beta(t) \leq c e^{\int_0^t r(\tau) d\tau} \quad \forall t \in [0, T]$$

$r(\tau) > 0$. $\beta \in C^1$ fn on some interval $[0, T]$.

Proof: Multiply by integration factor to solve: $G(t) = c + \int_0^t r(\tau) \beta(\tau) d\tau$

what is $\frac{dG(t)}{dt}$

Going back to $\textcircled{A}$, multiply by $e^{+\alpha t}$.

$$e^{\alpha t} \|y(t)\| \leq C \|y(0)\| + CK \int_0^t e^{+\alpha \tau} \|y(\tau)\| d\tau$$

$$\beta(t) \equiv e^{\alpha t} \|y(t)\|$$

$$\beta(t) \leq C \|y(0)\| + CK \int_0^t \beta(\tau) d\tau$$

$\Rightarrow$ By Gronwall's inequality,

$$\beta(t) \leq C e^{CKt} \|y(0)\|$$

$$e^{\alpha t} \|y(t)\| \leq C e^{CKt} \|y(0)\|$$

$$\|y(t)\| \leq C e^{(CK-\alpha)t} \|y(0)\|$$

$$CK - \alpha < 0$$

$$\text{if } K < \alpha/C, \|y(t)\| \rightarrow 0.$$

$$\|\phi^t(x) - x^*\| \rightarrow 0$$

$\alpha < -\text{Re}(\lambda)$ for all eigenvalues λ of A .

Holds for value of δ corresponding to $K < \alpha/C$

$$\phi^t(x) \rightarrow x^* \text{ as } t \rightarrow \infty.$$

as long as x is in a small nghd of x^* . (ie $\|y(0)\| = \|x - x^*\| < \delta$).

Hartman-Grobman theorem:

Around a hyperbolic fixed point (eigs of A do not have $\text{Re part } 0$), there exists a coordinate transformation

$$h: \mathbb{R}^n \rightarrow \mathbb{R}^n \text{ s.t.}$$

$$\phi^t = \underline{h}^{-1} \circ \underline{e}^{tA} \circ \underline{h}.$$